Assignment 1
IB3702 Mathematics for Machine learning

Harman Singh

October 3, 2025

Problem 1

Question 1.1

(a) Considering a_0 is 1, we have to simply fill in 1 where it says a_n in the formula. The result of this will be a_1. Then we will take a_1 and fill it in the formula to get a_2. We will repeat this process until we reach a_4.

- $a_0 = 1$
- $a_1 = \frac{1}{2}(1 + \frac{2}{1}) = \frac{1}{2} * 3 = \frac{3}{2}$ **So** $a_1 = \frac{3}{2}$
- $a_2 = \frac{1}{2}(\frac{3}{2} + \frac{2}{\frac{3}{2}}) = \frac{1}{2} * (\frac{3}{2} + \frac{4}{3}) = \frac{1}{2} * \frac{17}{6} = \frac{17}{12}$ **So** $a_2 = \frac{17}{12}$
- $a_3 = \frac{1}{2}(\frac{17}{12} + \frac{2}{\frac{17}{12}}) = \frac{1}{2} * (\frac{17}{12} + \frac{24}{17}) = \frac{1}{2} * \frac{577}{204} = \frac{577}{408}$ **So** $a_3 = \frac{577}{408}$
- $a_4 = \frac{1}{2}(\frac{577}{408} + \frac{2}{\frac{577}{408}}) = \frac{1}{2} * (\frac{577}{408} + \frac{816}{577}) = \frac{1}{2} * \frac{665{,}857}{235{,}416} = \frac{665{,}857}{470{,}832}$ **So** $a_4 = \frac{665{,}857}{470{,}832}$

(b) answer/solution to this problem

Question 1.2

I seperate this equation in two parts:

- Left part: $(-1)^n$
- Right part: $\frac{n}{n+1}$

For the left part, for even values of n, the result will always be 1, and for odd values of n, the result will always be -1

Then for the right part, as n increases, the result will get closer and closer to 1, but will never reach 1. So the limit of the right part is 1.

So, if we combine both parts, we can see that for even values of n, the result will be 1 * (something that approaches 1) which will approach 1. And for odd values of n, the result will be -1 * (something that approaches 1) which will approach -1.

This means that the sequence diverges, since it does not approach 1 single value.

Problem 2

Question 2.1

To know if this series converges or diverges, I will use the integral test. For this, I have to rewrite the series in function form (so it's easier for me to visualise) and then I convert it to integral form:

$$f(x) = \frac{1}{x(\ln x)} \to \int_2^\infty \frac{1}{x(\ln x)}dx$$

Then we do substituion:

- $u = \ln x$
- $du = \frac{1}{x}dx$
- $dx = x \cdot du$

$$\int_2^\infty \frac{1}{x(\ln x)}dx = \frac{1}{x \cdot u} \cdot x\ du = \frac{1}{u}du$$

initially the limits were from 2 to ∞, but now we need to change them to u values:

- when $x = 2$, $u = \ln 2$
- when $x \to \infty$, $u \to \infty$

So now the integral looks like this:

$$\int_{\ln 2}^\infty \frac{1}{u}du$$

And now we can solve the integral using the standard formula $\int \frac{1}{u}du = \ln|u| + C$:

$$\int_{\ln 2}^\infty \frac{1}{u}du = \ln|u|\Big|_{\ln 2}^\infty = \lim_{t\to\infty}[\ln(\ln t) - \ln(\ln 2)] = \infty - \ln(\ln 2) = \infty$$

Since the result of the integral is ∞ (the integral diverges), that means the series also diverges.

Question 2.2

$$\sum_{n=0}^\infty (-1)^n (\frac{1}{e})^n$$

You can rewrite geometric series as:

$$\sum_{n=0}^{\infty} r^n \quad \text{with} \quad |r| = -\frac{1}{e}$$

A geometric series converges if $|r| < 1$. In this case, $|r| = \frac{1}{e}$, and since e is approximately 2.718, this series converges because $\frac{1}{\text{something larger than 1}}$ is < 1

To know what it converges to, I use the formula $\frac{a}{1-r}$, where a is the first term of the series.

- $a = 1$
- $r = -\frac{1}{e}$

$$\frac{1}{1-(-\frac{1}{e})} = \frac{1}{1+\frac{1}{e}} = \frac{1}{\frac{e+1}{e}} = \frac{e}{e+1}$$

Problem 3

To answer the following questions, I will look at the limits on both sides

Question 1

- for $x < 0$ we need to look at the first line $\rightarrow$ the limit is 1
- for $x > 0$ we need to look at the second line $\rightarrow$ the limit is 0

So, no, the function can not be defined at $x = 0$ to make it continuous, since the limits on both sides are not equal.

Question 2

- for $x < 1$ we need to look at the second line $\rightarrow$ the limit is 1
- for $x > 1$ we need to look at the third line $\rightarrow$ the limit is $2 - x$, so also 1

So, yes, the function can be defined at $x = 1$ to make it continuous, since the limits on both sides are equal. The value of the function at $x = 1$ will be 1. So we set $f(1) = 1$.

Question 3

- for $x < 3$ we need to look at the third line $\rightarrow$ the limit is $2 - x$, so -1
- for $x > 3$ we need to look at the fourth line $\rightarrow$ the limit is $x - 4$ so also -1

So, yes, the function can be defined at $x = 3$ to make it continuous, since the limits on both sides are equal. The value of the function at $x = 3$ will be -1. So we set $f(3) = -1$.

Problem 4

Question 4.1

(1) So first and foremost: $\sin^2(x) = (\sin(x))^2$, so we can rewrite the function as:

$$f(x) \equiv \sin(\ln(x^2+1))^2$$

Then we just use the chain rule: $\frac{d}{dx}f(g(x))' = f'(g(x)) \cdot g'(x)$ to get:

$$f'(x) = 2 \cdot \sin(\ln(x^2+1)) \cdot \cos(\ln(x^2+1)) \cdot \frac{d}{dx}(\ln(x^2+1))$$

Now I just need to calculate $\frac{d}{dx}(\ln(x^2+1))$:

$$\frac{d}{dx}(\ln(x^2+1)) = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}$$

So the final answer is:

$$f'(x) = 2 \cdot \sin(\ln(x^2+1)) \cdot \cos(\ln(x^2+1)) \cdot \frac{2x}{x^2+1}$$

---

(2) So here I start with the left part first: $2^{\cos x}$. I will use the rule: $\frac{d}{dx}a^{f(x)} = a^{f(x)} \cdot \ln(a) \cdot f'(x)$. So we get:

$$\frac{d}{dx}2^{\cos x} = 2^{\cos x} \cdot \ln(2) \cdot (-\sin x) = -\ln(2) \cdot \sin x \cdot 2^{\cos x}$$

Then the right part.
$\sqrt{x^3}$ can be rewritten as $x^{3/2}$, so we get $\frac{x^2}{x^{3/2}}$. And this can be rewritten again as $x^{2-3/2} = x^{1/2} = \sqrt{x}$. Now we can calculate the derivative:

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$$

So all together, we get:

$$g'(x) = -\ln(2)\sin x \cdot 2^{\cos x} + \frac{1}{2\sqrt{x}}$$

Question 4.2

For these questions, I seperate the function in two parts, calculate the partial derivative of both parts, and then add them together. The centered function is the final answer for that question.

(1) $\frac{\partial f}{\partial x}$

I will treat y as a constant, and just differentiate with respect to x.

- left side: $x\sin(y)$
- right side: e^{x+2y}

LEFT: $\frac{\partial}{\partial x}(x\sin(y)) = \sin(y)$

RIGHT: $\frac{\partial}{\partial x}(e^{x+2y}) = e^{x+2y} \cdot \frac{\partial}{\partial x}(x+2y) = e^{x+2y} \cdot \frac{\partial}{\partial x}(1+0) = e^{x+2y} \cdot \frac{\partial}{\partial x}(1) = e^{x+2y} \cdot 1$
$= e^{x+2y}$

$$\frac{\partial f}{\partial x} = \sin(y) + e^{x+2y}$$

(2) $\frac{\partial f}{\partial y}$

Here I will treat x as a constant, and just differentiate with respect to y.

LEFT: since x is constant,

$\frac{\partial}{\partial x}(x\sin(y)) = x \cdot \frac{\partial}{\partial y}(\sin(y)) = x \cdot \cos(y)$

RIGHT: for the right side we just the chain rule again

$\frac{\partial}{\partial x}(e^{x+2y}) = e^{x+2y} \cdot \frac{\partial}{\partial y}(x+2y) = e^{x+2y} \cdot \frac{\partial}{\partial y}(0+2) = e^{x+2y} \cdot \frac{\partial}{\partial y}(2) = e^{x+2y} \cdot 0 + e^{x+2y} \cdot 1$
$= 2e^{x+2y}$

$$\frac{\partial f}{\partial y} = x\cos(y) + 2e^{x+2y}$$

(3) $\frac{\partial^2 f}{\partial x \partial y}$

For this one I will just take what we found in (2) and differentiate it with respect to x.

LEFT:

$\frac{\partial}{\partial x}(x\cos(y)) = \cos(y)$

RIGHT:

$\frac{\partial}{\partial x}(2e^{x+2y}) = 2e^{x+2y} \cdot \frac{\partial}{\partial x}(x+2y) = 2e^{x+2y} \cdot \frac{\partial}{\partial x}(1+0) = 2e^{x+2y} \cdot \frac{\partial}{\partial x}(1) = 2e^{x+2y} \cdot 1 = 2e^{x+2y}$

$$\frac{\partial^2 f}{\partial x \partial y} = \cos(y) + 2e^{x+2y}$$

(4) $\frac{\partial^2 f}{\partial x^2}$

For this one I will just take what we found in (1) and differentiate it with respect to x again.

LEFT: this will just be 0, since y is constant

$\frac{\partial}{\partial x}(\sin(y)) = \frac{\partial}{\partial x}(\sin(0)) = \frac{\partial}{\partial x}(0) = 0$

RIGHT:

$\frac{\partial}{\partial x}(e^{x+2y}) = e^{x+2y} \cdot \frac{\partial}{\partial x}(x+2y) = e^{x+2y} \cdot \frac{\partial}{\partial x}(1+0) = e^{x+2y} \cdot \frac{\partial}{\partial x}(1) = e^{x+2y} \cdot 1 = e^{x+2y}$

$$\frac{\partial^2 f}{\partial x^2} = 0 + e^{x+2y} = e^{x+2y}$$

(5) $\frac{\partial^2 f}{\partial y \partial x}$

For this one I will just take what we found in (1) and differentiate it with respect to y.

LEFT:

$\frac{\partial}{\partial y}(\sin(y)) = \cos(y)$

RIGHT:

$\frac{\partial}{\partial y}(e^{x+2y}) = e^{x+2y} \cdot \frac{\partial}{\partial y}(x+2y) = e^{x+2y} \cdot \frac{\partial}{\partial y}(0+2) = e^{x+2y} \cdot \frac{\partial}{\partial y}(2) = e^{x+2y} \cdot 0 + e^{x+2y} \cdot 1$
$= 2e^{x+2y}$

$$\frac{\partial^2 f}{\partial y \partial x} = \cos(y) + 2e^{x+2y}$$

Problem 5

the domain

The domain of a function is the set of all possible input values (x-values) that will produce a valid output for the function.

Because this function is a fraction, and since division by zero is undefined, we just need to find the values of x that make the denominator equal to zero, and exclude those from the domain.

We need to solve the equation $x^2 - x - 2 = 0$. I factor this to $(x-2)(x+1) = 0$, and this means the solutions of the equations are $x = 2$ and $x = -1$. So now we just write the domain, which is all real numbers except 2 and -1:

$$(-\infty, -1) \cup (-1, 2) \cup (2, \infty)$$

the zeroes (also called roots)

So here we have to do the opposite. We now need to find the values of x that make the whole function equal to zero.

Because this function is a fraction, the result will be 0 if the numerator is 0 (as long as the denominator is not also 0).

We need to solve the equation $x^2 = 0$. The only solution to this equation is $x = 0$. So now we need to make sure the denominator is not also 0 at this point. If we fill in 0 in the denominator, we get $0^2 - 0 - 2 = -2$, which is not 0. So this function has 1 zero at:

$$x = 0$$

the horizontal and vertical asymptotes

For this I looked at the graph (next page), and I can see that there are 2 vertical asymptotes and 1 horizontal asymptote.

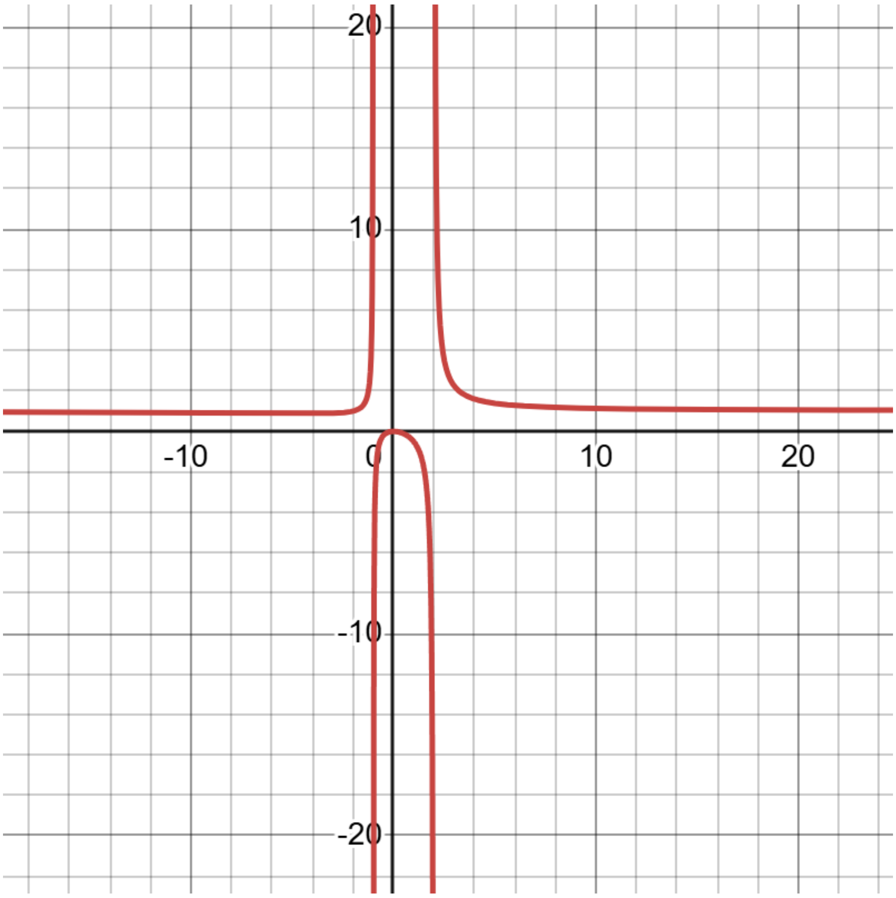
- Vertical: $x = -1$ and $x = 2$
- Horizontal: $y = 1$

the extreme values

graph

$$f(x) = \frac{x^2}{x^2 - x - 2}$$

Every line is 2 units apart. So for example, for the vertical assymptotes, you can see them at $x = -1$ and $x = 2$. And for the horizontal assymptote you can see it at $y = 1$



Problem 6

(1)

$$\int_1^2 (\frac{2+3x^2}{x} dx)$$

First I split the function in two parts

$$\int_1^2 (\frac{2}{x})dx + \int_1^2 (3x)dx$$

And now solve both parts:

$$\int_1^2 (\frac{2}{x} dx) = 2\ln|x|\Big|_1^2 = 2\ln(2) - 2\ln(1) = 2\ln(2) - 0 = 2\ln(2)$$

$$\int_1^2 (3x)dx = \frac{3x^2}{2}\Big|_1^2 = \frac{3(2^2)}{2} - \frac{3(1^2)}{2} = \frac{3(4)}{2} - \frac{3(1)}{2} = \frac{12}{2} - \frac{3}{2} = \frac{9}{2}$$

So together

$$2\ln(2) + \frac{9}{2}$$

(b)

$$\int x \cdot e^x dx$$

The formula for integration by parts is: $\int udv = uv - \int vdu$. So I need to choose u and dv. I choose:

- $u = x \to du = dx$
- $dv = e^x dx \to v = e^x$

Then I fill in the formula:

$$\int xe^x dx = xe^x - \int e^x dx$$

The integral of e^x is just e^x, so we get (and add the constante C):

$$= xe^x - e^x + C$$